

$$f\left(\frac{3}{2}\right) = \left(2 \times \frac{3}{2} - 3\right) \left(3 \times \frac{3}{2} + 5\right) Q\left(\frac{3}{2}\right) + a \times \frac{3}{2} + b$$

$$= \frac{3}{2}a + b$$

$$f\left(-\frac{5}{3}\right) = \left(2 \times \left(-\frac{5}{3}\right) - 3\right) \left(3 \times \left(-\frac{5}{3}\right) + 5\right) Q\left(-\frac{5}{3}\right) + a \times \left(-\frac{5}{3}\right) + b$$

$$= -\frac{5}{3}a + b$$

由題意知 $f\left(\frac{3}{2}\right) = 27$, $f\left(-\frac{5}{3}\right) = 8$

$$\text{故} \begin{cases} \frac{3}{2}a + b = 27 \cdots ① \\ -\frac{5}{3}a + b = 8 \cdots ② \end{cases}$$

$$① - ② \Rightarrow \frac{19}{6}a = 19 \Rightarrow a = 6$$

$$a = 6 \text{ 代入 } ① \Rightarrow \frac{3}{2} \times 6 + b = 27 \Rightarrow b = 18$$

因此 $a + b = 6 + 18 = 24$

類題 2

$\because f(x)$ 除以 $x-1$ 的商式為 $g(x)$, 餘數為 3

$$\therefore f(x) = (x-1) \times g(x) + 3$$

$\because g(x)$ 除以 $x-2$ 的餘數為 6 $\therefore g(2) = 6$

而 $f(x)$ 除以 $x-2$ 的餘數為

$$f(2) = (2-1) \times g(2) + 3 = 1 \times 6 + 3 = 9$$

範例 2

類題 1

$$\text{令 } f(x) = x^3 + ax^2 + 3x + b + 1$$

$\because x^2 + x - 2$ 是 $f(x)$ 的因式

$$\text{且 } x^2 + x - 2 = (x-1)(x+2)$$

$\therefore x-1$ 與 $x+2$ 也是 $f(x)$ 的因式

由因式定理知

$$f(1) = 1 + a + 3 + b + 1 = 0$$

$$\Rightarrow a + b = -5 \cdots ①$$

$$f(-2) = -8 + 4a - 6 + b + 1 = 0$$

$$\Rightarrow 4a + b = 13 \cdots ②$$

$$② - ① \Rightarrow 3a = 18 \Rightarrow a = 6$$

$$a = 6 \text{ 代入 } ① \Rightarrow 6 + b = -5 \Rightarrow b = -11$$

$$\text{因此 } a - b = 6 - (-11) = 17$$

類題 2

$$\text{令 } f(x) = ax^3 + 3x^2 + bx - 2$$

$\because x^2 - 3x + 2$ 是 $f(x)$ 的因式

$$\text{且 } x^2 - 3x + 2 = (x-1)(x-2)$$

$\therefore x-1$ 與 $x-2$ 也是 $f(x)$ 的因式

由因式定理知

$$f(1) = a + 3 + b - 2 = 0 \Rightarrow a + b = -1 \cdots ①$$

$$f(2) = 8a + 12 + 2b - 2 = 0 \Rightarrow 4a + b = -5 \cdots ②$$

$$② - ① \Rightarrow 3a = -4 \Rightarrow a = -\frac{4}{3}$$

$$a = -\frac{4}{3} \text{ 代入 } ① \Rightarrow -\frac{4}{3} + b = -1 \Rightarrow b = \frac{1}{3}$$

$$\text{因此 } a + 3b = -\frac{4}{3} + 3 \times \frac{1}{3} = -\frac{1}{3}$$

範例 3

類題 1

原式等號的兩側同乘以 $(x-1)(x+2)^2$

$$\Rightarrow 3x^2 + 9x - 3$$

$$= A(x+2)^2 + B(x-1)(x+2) + C(x-1)$$

$$= A(x^2 + 4x + 4) + B(x^2 + x - 2) + C(x-1)$$

$$= (A+B)x^2 + (4A+B+C)x + (4A-2B-C)$$

$$\begin{cases} A+B=3 \cdots ① \end{cases}$$

$$\text{則} \begin{cases} 4A+B+C=9 \cdots ② \end{cases}$$

$$\begin{cases} 4A-2B-C=-3 \cdots ③ \end{cases}$$

$$② + ③ \Rightarrow 8A - B = 6 \cdots ④$$

$$① + ④ \Rightarrow 9A = 9 \Rightarrow A = 1$$

$$A = 1 \text{ 代入 } ① \Rightarrow 1 + B = 3 \Rightarrow B = 2$$

類題 2

原式等號的兩側同乘以 $(2x-3)^2$

$$\Rightarrow 4x^2 - 6x - 3$$

$$= a(2x-3)^2 + b(2x-3) + c$$

$$= a(4x^2 - 12x + 9) + b(2x-3) + c$$

$$= 4ax^2 + (-12a+2b)x + (9a-3b+c)$$

$$\begin{cases} 4a = 4 \cdots ① \end{cases}$$

$$\text{則} \begin{cases} -12a + 2b = -6 \cdots ② \end{cases}$$

$$\begin{cases} 9a - 3b + c = -3 \cdots ③ \end{cases}$$

由 ① 得 $a = 1$

$$a = 1 \text{ 代入 } ② \Rightarrow -12 \times 1 + 2b = -6 \Rightarrow b = 3$$

$$a = 1, b = 3 \text{ 代入 } ③ \Rightarrow 9 \times 1 - 3 \times 3 + c = -3$$

$$\Rightarrow c = -3$$

$$\text{故 } a + b + 2c = 1 + 3 + 2 \times (-3) = -2$$

歷屆試題

P.92

- | | | | |
|---------|---------|---------|---------|
| 1. (A) | 2. (B) | 3. (B) | 4. (D) |
| 5. (D) | 6. (D) | 7. (B) | 8. (A) |
| 9. (D) | 10. (D) | 11. (C) | 12. (D) |
| 13. (A) | 14. (A) | 15. (D) | 16. (C) |
| 17. (C) | 18. (C) | | |

解析：

$$1. f(x) = 3x^3 - 7x^2 + 4x - 6$$

$$= d(x-2)^3 + c(x-2)^2 + b(x-2) + a$$

$$\begin{array}{r} 3 \quad -7 \quad +4 \quad -6 \quad | \quad 2 \\ +6 \quad -2 \quad +4 \quad | \\ \hline 3 \quad -1 \quad +2 \quad | \end{array} \begin{array}{l} -2 \rightarrow a \\ +6 \quad +10 \\ \hline 3 \quad +5 \quad | \end{array} \begin{array}{l} 12 \rightarrow b \\ +6 \\ \hline 3 \quad +5 \quad | \end{array} \begin{array}{l} 11 \rightarrow c \\ +6 \\ \hline 3 \quad +5 \quad | \end{array}$$

$$d \leftarrow 3, \underline{11} \rightarrow c$$

$$\text{故 } a = -2, b = 12, c = 11, d = 3$$

$$\text{因此 } a - b - c - d = -2 - 12 - 11 - 3 = -28$$

$$2. \text{ 設 } f(x) \text{ 除以 } (x-1)(x^2+x+1) \text{ 的商式為 } Q(x) \text{ 則}$$

$$f(x) \div [(x-1)(x^2+x+1)] = Q(x) \cdots (3x^2+5x-2)$$

由除法原理

$$f(x) = [(x-1)(x^2+x+1)]Q(x) + (3x^2+5x-2)$$

依上式可知， $f(x)$ 除以 x^2+x+1 的餘式就是 $3x^2+5x-2$ 除以 x^2+x+1 的餘式

$$\begin{array}{r} 3 \\ 1+1+1 \overline{) 3+5-2} \quad , \text{ 故所求餘式為 } 2x-5 \\ \underline{3+3+3} \\ 2-5 \end{array}$$

$$3. (1) \because ax^4 + bx^3 + 6x^2 + 5x + 2 \text{ 除以 } (x+1)^2 \text{ 的餘式為 } 3x+4$$

$$\therefore (ax^4 + bx^3 + 6x^2 + 5x + 2) - (3x+4)$$

$$= ax^4 + bx^3 + 6x^2 + 2x - 2$$

可以被 $(x+1)^2$ 整除

$$(2) (x+1)^2 = x^2 + 2x + 1$$

則 $ax^4 + bx^3 + 6x^2 + 2x - 2$ 除以 $(x+1)^2$ 的算式如下：

$$\begin{array}{r} a + (b-2a) - \quad \quad \quad 2 \\ 1+2+1 \overline{) a + \quad b + \quad \quad 6 + \quad \quad 2 - 2} \\ \underline{a + \quad 2a + \quad \quad a} \\ (b-2a) + \quad (6-a) + \quad \quad 2 \\ \underline{(b-2a) + \quad (2b-4a) + \quad (b-2a)} \\ (3a-2b+6) + (2a-b+2) - 2 \\ \underline{\quad \quad \quad - \quad \quad 2 \quad \quad - \quad \quad 4 \quad - 2} \\ (3a-2b+8) + (2a-b+6) + 0 \end{array}$$

$$\text{則 } 3a - 2b + 8 = 0 \text{ 且 } 2a - b + 6 = 0$$

$$\Rightarrow \begin{cases} 3a - 2b = -8 \cdots \cdots ① \\ 2a - b = -6 \cdots \cdots ② \end{cases}$$

$$① - ② \times 2 : -a = 4 \Rightarrow a = -4$$

$$a = -4 \text{ 代入 } ② :$$

$$2 \times (-4) - b = -6 \Rightarrow b = -2$$

$$\text{因此 } a + b = -4 + (-2) = -6$$

$$4. \frac{1}{\sqrt{2}+1} = \frac{1 \times (\sqrt{2}-1)}{(\sqrt{2}+1) \times (\sqrt{2}-1)} = \frac{\sqrt{2}-1}{(\sqrt{2})^2 - 1^2} = \sqrt{2}-1$$

$$\therefore \text{ 立方差 : } (x-y)(x^2+xy+y^2) = x^3 - y^3$$

$$\begin{aligned} \therefore \left(\frac{1}{\sqrt{2}+1} - 1 \right) \left[\left(\frac{1}{\sqrt{2}+1} \right)^2 + \frac{1}{\sqrt{2}+1} + 1 \right] \\ = \left(\frac{1}{\sqrt{2}+1} \right)^3 - 1^3 = (\sqrt{2}-1)^3 - 1 \\ = \left[(\sqrt{2})^3 - 3 \times (\sqrt{2})^2 \times 1 + 3 \times \sqrt{2} \times 1^2 - 1^3 \right] - 1 \\ = (2\sqrt{2} - 6 + 3\sqrt{2} - 1) - 1 = (5\sqrt{2} - 7) - 1 \\ = -8 + 5\sqrt{2} \end{aligned}$$

$$5. \text{ 設 } f(x) \div (x^2 - 3x + 2) = Q(x) \cdots (3x - 4)$$

$$\text{則 } f(x) = (x^2 - 3x + 2) \times Q(x) + (3x - 4)$$

$$\text{而 } f(1) = (1^2 - 3 \times 1 + 2) \times Q(1) + (3 \times 1 - 4)$$

$$= 0 \times Q(1) + (-1) = -1$$

$$\text{又 } x-1 \text{ 除 } g(x) \text{ 所得餘式為 } 5 \Rightarrow g(1) = 5$$

故以 $x-1$ 除 $f(x) + g(x)$ 的餘式為

$$f(1) + g(1) = -1 + 5 = 4$$

$$6. (A) \text{ 令 } x+1=0 \Rightarrow x=-1$$

$$f(-1) = 2 \times (-1)^3 + (-1)^2 - 7 \times (-1) - 6 = 0$$

$$(B) \text{ 令 } x-2=0 \Rightarrow x=2$$

$$f(2) = 2 \times 2^3 + 2^2 - 7 \times 2 - 6 = 0$$

$$(C) \text{ 令 } 2x+3=0 \Rightarrow x=-\frac{3}{2}$$

$$\begin{aligned} f\left(-\frac{3}{2}\right) &= 2 \times \left(-\frac{3}{2}\right)^3 + \left(-\frac{3}{2}\right)^2 - 7 \times \left(-\frac{3}{2}\right) - 6 \\ &= 0 \end{aligned}$$

$$(D) \text{ 令 } 2x-1=0 \Rightarrow x=\frac{1}{2}$$

$$f\left(\frac{1}{2}\right) = 2 \times \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^2 - 7 \times \left(\frac{1}{2}\right) - 6 = -9 \neq 0$$

因此 $x+1$ 、 $x-2$ 、 $2x+3$ 都是 $f(x)$ 的因式，

而 $2x-1$ 不是 $f(x)$ 的因式

$$\begin{aligned} 7. (1) \because (\sqrt{3} \pm i)^2 &= (\sqrt{3})^2 \pm 2 \times \sqrt{3} \times i + i^2 \\ &= 3 \pm 2\sqrt{3}i + (-1) = 2 \pm 2\sqrt{3}i \end{aligned}$$

$$\begin{aligned} \therefore \frac{(\sqrt{3}-i)^2}{(\sqrt{3}+i)^2} &= \frac{(\sqrt{3}-i)^2}{(\sqrt{3}+i)^2} = \frac{2-2\sqrt{3}i}{2+2\sqrt{3}i} \\ &= \frac{2 \times (1-\sqrt{3}i)}{2 \times (1+\sqrt{3}i)} = \frac{1-\sqrt{3}i}{1+\sqrt{3}i} \\ &= \frac{(1-\sqrt{3}i)^2}{(1+\sqrt{3}i)(1-\sqrt{3}i)} \end{aligned}$$

$$\begin{aligned}
 &= \frac{1^2 - 2 \times 1 \times \sqrt{3}i + (\sqrt{3}i)^2}{1^2 + (\sqrt{3})^2} \\
 &= \frac{1 - 2\sqrt{3}i + (-3)}{4} = \frac{-2 - 2\sqrt{3}i}{4} \\
 &= \frac{2 \times (-1 - \sqrt{3}i)}{4} = \frac{-1 - \sqrt{3}i}{2}
 \end{aligned}$$

$$(2) \because \left(\frac{\sqrt{3}-i}{\sqrt{3}+i} \right)^2 \text{ 與 } \left(\frac{\sqrt{3}+i}{\sqrt{3}-i} \right)^2 \text{ 互為倒數}$$

$$\begin{aligned}
 \therefore \left(\frac{\sqrt{3}+i}{\sqrt{3}-i} \right)^2 &= \frac{2}{-1-\sqrt{3}i} \\
 &= \frac{2(-1+\sqrt{3}i)}{(-1-\sqrt{3}i)(-1+\sqrt{3}i)} = \frac{2(-1+\sqrt{3}i)}{(-1)^2 + (\sqrt{3})^2} \\
 &= \frac{2(-1+\sqrt{3}i)}{4} = \frac{-1+\sqrt{3}i}{2}
 \end{aligned}$$

由(1)和(2)可知

$$\begin{aligned}
 \left(\frac{\sqrt{3}-i}{\sqrt{3}+i} \right)^2 + \left(\frac{\sqrt{3}+i}{\sqrt{3}-i} \right)^2 &= \frac{-1-\sqrt{3}i}{2} + \frac{-1+\sqrt{3}i}{2} \\
 &= -1 = -1 + 0i
 \end{aligned}$$

則 $a = -1$, $b = 0$, 因此 $a + b = -1 + 0 = -1$

8. 在方程式 $4x^2 - 2x - 5 = 0$ 之中

令 $a = 4$, $b = -2$, $c = -5$

$\therefore$ 方程式有兩根 α , β

$\therefore$ 兩根積 $\alpha\beta = \frac{c}{a} = \frac{-5}{4}$, 故選(A)

9. (1) $\because a, b$ 為實數且 $a, b+2i, -1+ai$ 為實係數三次方程式 $f(x) = 0$ 之三根

$\therefore b+2i$ 與 $-1+ai$ 互為共軛複數

而 $b+2i$ 的共軛複數 $\overline{b+2i} = b-2i$

則 $b-2i = -1+ai \Rightarrow b = -1, a = -2$

因此 $f(x) = 0$ 的三根為 $-2, -1 \pm 2i$

(2) 先求以 $-1 \pm 2i$ 為兩根的二次方程式:

令 $x = -1 \pm 2i$

$$\text{則 } x+1 = \pm 2i \Rightarrow (x+1)^2 = (\pm 2i)^2$$

$$\Rightarrow x^2 + 2x + 1 = -4 \Rightarrow x^2 + 2x + 5 = 0$$

因此 $x^2 + 2x + 5$ 是 $f(x)$ 的因式

$$(3) \because [x - (-2)](x^2 + 2x + 5)$$

$$= (x+2)(x^2 + 2x + 5)$$

$$= x^3 + 4x^2 + 9x + 10$$

$$\therefore f(x) = k(x^3 + 4x^2 + 9x + 10)$$

其中 $k \neq 0$

故選(D)

$$\begin{aligned}
 10. (1+2i)(2-i) &= 2-i+4i-2i^2 \\
 &= 2-i+4i-(-2) = 4+3i
 \end{aligned}$$

則 $a+bi = 4+3i$ (a, b 為實數)

$\Rightarrow a = 4, b = 3$, 故 $a+b = 4+3 = 7$

$$11. \text{ 方程式 } x^2 + kx = k-3 \Rightarrow x^2 + kx + (-k+3) = 0$$

$\because$ 方程式有兩個共軛虛根 $\therefore$ 其判別式 < 0

$$\text{即 } k^2 - 4 \times 1 \times (-k+3) < 0$$

$$\Rightarrow k^2 + 4k - 12 < 0 \Rightarrow (k+6)(k-2) < 0$$

$$\Rightarrow -6 < k < 2$$

則 k 之解的範圍區間 $(a, b) = (-6, 2)$

故 $a = -6, b = 2$, 因此 $b-a = 2 - (-6) = 8$

$$12. \frac{3x-1}{(x-3)(x-1)} = \frac{A}{x-3} + \frac{B}{x-1}$$

上式等號的兩側同乘以 $(x-3)(x-1)$, 得

$$3x-1 = A(x-1) + B(x-3) \cdots \cdots \textcircled{1}$$

(1) 令 $x = 3$ 代入 $\textcircled{1}$:

$$3 \times 3 - 1 = A \times (3-1) + B \times (3-3)$$

$$\Rightarrow 8 = 2A + 0 \Rightarrow A = 4$$

(2) 令 $x = 1$ 代入 $\textcircled{1}$:

$$3 \times 1 - 1 = A \times (1-1) + B \times (1-3)$$

$$\Rightarrow 2 = 0 - 2B \Rightarrow B = -1$$

由(1)和(2)可知, $A = 4, B = -1$, 故選(D)

[另解]

$$\frac{3x-1}{(x-3)(x-1)} = \frac{A}{x-3} + \frac{B}{x-1}$$

上式等號的兩側同乘以 $(x-3)(x-1)$, 得

$$\begin{aligned}
 3x-1 &= A(x-1) + B(x-3) = (Ax-A) + (Bx-3B) \\
 &= (A+B)x + (-A-3B)
 \end{aligned}$$

$$\text{由多項式相等, 則 } \begin{cases} A+B=3 \cdots \cdots \textcircled{1} \\ -A-3B=-1 \cdots \cdots \textcircled{2} \end{cases}$$

$$\textcircled{1} + \textcircled{2}: -2B = 2 \Rightarrow B = -1$$

$$B = -1 \text{ 代入 } \textcircled{1}: A + (-1) = 3 \Rightarrow A = 4$$

因此 $A = 4, B = -1$, 故選(D)

13. [解1]

$$\frac{x^2 + 2x + 7}{(x-2)(x+2)(x+3)} = \frac{A}{x-2} + \frac{B}{x+2} + \frac{C}{x+3}$$

上式等號的兩側同乘以 $(x-2)(x+2)(x+3)$, 得

$$x^2 + 2x + 7 = A(x+2)(x+3) + B(x-2)(x+3)$$

$$+ C(x-2)(x+2)$$

$$= A(x^2 + 5x + 6) + B(x^2 + x - 6)$$

$$+ C(x^2 - 4)$$

$$= (Ax^2 + 5Ax + 6A) + (Bx^2 + Bx - 6B)$$

$$+ (Cx^2 - 4C)$$

$$= (A+B+C)x^2 + (5A+B)x + (6A-6B-4C)$$

則 x^2 項的係數： $A+B+C=1$

〔解2〕

$$\frac{x^2+2x+7}{(x-2)(x+2)(x+3)} = \frac{A}{x-2} + \frac{B}{x+2} + \frac{C}{x+3}$$

上式等號的兩側同乘以 $(x-2)(x+2)(x+3)$ ，得

$$x^2+2x+7 = A(x+2)(x+3) + B(x-2)(x+3) + C(x-2)(x+2) \cdots \cdots ①$$

(1) 令 $x=2$ 代入①：

$$2^2+2 \times 2+7 = A(2+2)(2+3) + B(2-2)(2+3) + C(2-2)(2+2)$$

$$\Rightarrow 15 = 20A \Rightarrow A = \frac{3}{4}$$

(2) 令 $x=-2$ 代入①：

$$\begin{aligned} (-2)^2+2 \times (-2)+7 &= A(-2+2)(-2+3) \\ &\quad + B(-2-2)(-2+3) \\ &\quad + C(-2-2)(-2+2) \end{aligned}$$

$$\Rightarrow 7 = -4B \Rightarrow B = -\frac{7}{4}$$

(3) 令 $x=-3$ 代入①：

$$\begin{aligned} (-3)^2+2 \times (-3)+7 &= A(-3+2)(-3+3) \\ &\quad + B(-3-2)(-3+3) \\ &\quad + C(-3-2)(-3+2) \end{aligned}$$

$$\Rightarrow 10 = 5C \Rightarrow C = 2$$

由(1)、(2)、(3)可知， $A+B+C = \frac{3}{4} + \left(-\frac{7}{4}\right) + 2 = 1$

14. 對任意實數 x

$$(1) \sqrt[3]{x^3} = x, \sqrt{x^2} = |x|$$

而 $|x|$ 和 x 不一定相等

(2) 當 $2+x \geq 0$ 時

$$\sqrt{(2+x)^2} = |2+x| = 2+x$$

$$\text{而 } (2+x)^2 = 4+4x+x^2$$

$$\text{則 } \sqrt{4+4x+x^2} = 2+x$$

$$(3) \sqrt[3]{(2-x)^3} = 2-x$$

$$\text{而 } (2-x)^3 = 8-12x+6x^2-x^3$$

$$\text{則 } \sqrt[3]{8-12x+6x^2-x^3} = 2-x$$

由(1)、(2)和(3)可知，選項(A)才是正確的

15. 〔解1〕

$$\frac{5}{(2x+1)(x-2)} = \frac{A}{2x+1} + \frac{B}{x-2}$$

將上式等號兩側同乘以 $(2x+1)(x-2)$

$$\text{可得 } 5 = A(x-2) + B(2x+1) \cdots \cdots ①$$

(1) 令 $x=2$ 代入①：

$$5 = A \times (2-2) + B \times (2 \times 2+1)$$

$$\Rightarrow 5 = 0 + 5B \Rightarrow 5B = 5 \Rightarrow B = 1$$

(2) 令 $x=-\frac{1}{2}$ 代入①：

$$5 = A \times \left(-\frac{1}{2}-2\right) + B \times \left(2 \times \left(-\frac{1}{2}\right)+1\right)$$

$$\Rightarrow 5 = -\frac{5}{2}A + 0 \Rightarrow A = -2$$

由(1)和(2)可知

$$3A+2B = 3 \times (-2) + 2 \times 1 = -6+2 = -4$$

〔解2〕

$$\frac{5}{(2x+1)(x-2)} = \frac{A}{2x+1} + \frac{B}{x-2}$$

將上式等號兩側同乘以 $(2x+1)(x-2)$

$$\text{可得 } 5 = A(x-2) + B(2x+1)$$

$$= (Ax-2A) + (2Bx+B)$$

$$= (A+2B)x + (-2A+B)$$

由多項式相等，則 $\begin{cases} A+2B=0 \cdots \cdots ① \\ -2A+B=5 \cdots \cdots ② \end{cases}$

$$① \times 2 : 2A+4B=0 \cdots \cdots ③$$

$$② + ③ : 5B=5 \Rightarrow B=1$$

$$\text{將 } B=1 \text{ 代入 } ① : A+2 \times 1=0 \Rightarrow A=-2$$

$$\text{因此 } 3A+2B = 3 \times (-2) + 2 \times 1 = -6+2 = -4$$

16. 設 $x^2+(a-5)x+(a+3)=0$ 的兩根為 α 、 β

$$(1) \text{ 兩根和： } \alpha+\beta = -\frac{a-5}{1} = -a+5$$

$\therefore$ 兩根均為正根

$$\therefore -a+5 > 0 \Rightarrow a < 5$$

$$(2) \text{ 兩根積： } \alpha\beta = \frac{a+3}{1} = a+3$$

$\therefore$ 兩根均為正根

$$\therefore a+3 > 0 \Rightarrow a > -3$$

(3) 方程式的判別式：

$$(a-5)^2 - 4 \times 1 \times (a+3)$$

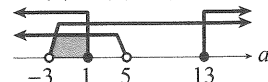
$$= a^2 - 14a + 13 = (a-1)(a-13)$$

$\therefore$ 方程式的兩根均為正根

$$\therefore \text{判別式 } (a-1)(a-13) \geq 0$$

$$\Rightarrow a \leq 1 \text{ 或 } a \geq 13$$

由(1)、(2)、(3)知， $-3 < a \leq 1$



而滿足 a 的實數解為 $m < a \leq n$

$$\text{故 } m = -3, n = 1, \text{ 因此 } m+n = -3+1 = -2$$

17. $\therefore \alpha, \beta$ 為方程式 $x^2 + 5x + k = 0$ 之兩根

$\therefore$ 兩根和 $\alpha + \beta = -\frac{5}{1} = -5$, 兩根積 $\alpha\beta = \frac{k}{1} = k$

$$\begin{aligned}\text{而 } \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta = (-5)^2 - 2 \times k \\ &= 25 - 2k\end{aligned}$$

$\therefore f(x) = 2x^2 + 7x + 5$ 除以 $x - \alpha$ 、 $x - \beta$ 的餘式分別為 -1 、 2

$$\therefore f(\alpha) = 2\alpha^2 + 7\alpha + 5 = -1 \cdots \textcircled{1}$$

$$f(\beta) = 2\beta^2 + 7\beta + 5 = 2 \cdots \textcircled{2}$$

$$\begin{aligned}\textcircled{1} + \textcircled{2} &\Rightarrow 2(\alpha^2 + \beta^2) + 7(\alpha + \beta) + 10 = 1 \\ &\Rightarrow 2 \times (25 - 2k) + 7 \times (-5) + 10 = 1 \\ &\Rightarrow 50 - 4k + (-35) + 10 = 1 \\ &\Rightarrow -4k = -24 \Rightarrow k = 6\end{aligned}$$

18. $f(x)$ 為三次多項式且領導係數為 a

$$\text{而 } f(1) = f(2) = f(-2) = 2$$

$$\begin{aligned}\text{設 } f(x) &= a(x-1)(x-2)[x-(-2)] + 2 \\ &= a(x-1)(x-2)(x+2) + 2\end{aligned}$$

$$\begin{aligned}f(-1) &= a(-1-1)(-1-2)(-1+2) + 2 \\ &= a \times (-2) \times (-3) \times 1 + 2 = 6a + 2\end{aligned}$$

$$\therefore f(-1) = 8 \quad \therefore 6a + 2 = 8 \Rightarrow a = 1$$

$$\begin{aligned}\text{則 } f(x) &= 1 \times (x-1)(x-2)(x+2) + 2 \\ &= (x-1)(x^2 - 4) + 2 \\ &= (x^3 - x^2 - 4x + 4) + 2 \\ &= x^3 - x^2 - 4x + 6\end{aligned}$$

與 $f(x) = ax^3 + bx^2 + cx + d$ 比較係數

得 $a = 1$, $b = -1$, $c = -4$, $d = 6$

故選(C)